Safia Fatima

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github.com/SafiaFatima | Google Scholar

Professional Summary

An experienced computer scientist with 8 years in AI, data science, and software engineering. Demonstrated strong problem-solving abilities through work in self-adaptive systems, reinforcement learning, LLMs, medical image processing, and bioinformatics. Committed to creating innovative solutions while continuously learning and adding value to company growth.

Technical Skills

Programming:	Python, SQL, Bash, C/C++, Java, HTML/CSS/JavaScript
AI & ML:	TensorFlow, Keras, PyTorch, Ray RLlib, OpenAI Gym, Hugging Face, Scikit-learn Architectures: RL, CNNs, GANs, LSTMs, Transformers
LLMs & Agents:	LangChain, LangGraph, Google ADK, MCP, Vertex AI, Pinecone, OpenTelemetry
Data & Analytics:	NumPy, Pandas, SciPy, Matplotlib, Seaborn, Plotly, Bayesian Statistics
Cloud & DevOps:	AWS, Azure, GCP, Docker, CI/CD, Git, REST APIs, Linting, Testing
Databases & MLOps:	PostgreSQL, MongoDB, SQLite, MLflow, Weights & Biases

Work Experience

Researcher & Developer, Simula Research Laboratory – Norway 03/2022 - Present

- Led doctoral research on "Towards Self-Healing Software Systems using Adaptive Learning"
- Proposed 3 novel scientific contributions in self-healing traffic systems, causality-based reinforcement learning, and LLM-based failure recovery for autonomous driving
- Gained extensive experience in simulated traffic system environments, reinforcement learning, attention networks, and large language models
- Applied software engineering best practices including version control, testing, and CI/CD while staying current with state-of-the-art AI advancements

Data Analyst, Odetta.ai – Remote 08/2021 - 01/2022

- Started to work part-time remotely as a data analyst and Python Scripter at a US-based company
- Working on projects related to data collection, data analytics, and web scraping

Lecturer & Lab Instructor, FAST NUCES – Pakistan 01/2018 - 01/2022

- Promoted from Lab Instructor to Lecturer under the Computer Science Department
- Taught courses including computer architecture, software design & analysis, object-oriented programming, Introduction to Computing, Databases, and Digital Logic Design
- Supervised CS undergraduate projects in the fields of computer vision, deep learning, and time series analysis

Education

University of Oslo, Ph.D. Computer Science – Norway 03/2022 - Present
FAST NUCES, M.Sc. Computer Science – Pakistan (CGPA: 3.44/4.0) 08/2016 - 07/2019
FAST NUCES, B.Sc. Computer Science – Pakistan (CGPA: 3.15/4.0) 08/2012 - 07/2016

Notable Projects

Hierarchical LLM-Based Failure Recovery for Autonomous Driving – GitHub | Paper

CRRL: A Causality-Based Reinforcement Learning Framework for the Autonomous System Recovery – GitHub | Paper

Self Healing of a Mixed Autonomy Traffic System Using Reinforcement Learning and Attention – GitHub | Paper

Evaluation of Multi-Modal Brain MRI Images for Glioblastomas – GitHub | Paper

Languages

English: Professional Working Proficiency (C2) **Urdu:** Professional Working Proficiency (C2) **Norwegian:** Elementary Proficiency (A1)